

Upper-bound companion to the combined score. The combined score of `new_approx_combined.txt` is derived as a lower bound on $\|Av\|_2^2$, where $A \in \mathbb{R}^{n \times d}$ is the full matrix, $M = A_B \in \mathbb{R}^{k \times d}$ is the block, $y := Mv$, $x := Av$, and $p_w := k/n$. This note derives a companion *upper* bound on $\|Av\|_2^2$ using the same two-ingredient template (deterministic L^p inequality plus an entropy-scaling assumption), and shows that the natural upper-bound counterpart is

$$\text{obj}_{\text{UB}}(v) := \frac{\|Mv\|_2^2 \cdot k_{\text{eff}}(v)}{p_w} = \frac{\|Mv\|_2^6}{p_w \|Mv\|_4^4}, \quad (1)$$

where $k_{\text{eff}}(v) := \|Mv\|_2^4 / \|Mv\|_4^4 = e^{h_2(y)}$ is the block participation ratio and $h_2(y) = -\log(\|y\|_4^4 / \|y\|_2^4)$ is the block collision (Rényi-2) entropy.

Template recap. The combined-score lower bound uses:

- (D1) Deterministic L^p inequality $\|y\|_4 \leq \|x\|_4$, from the fact that y is a subvector of x .
- (D2) Dimension-scaled entropy heuristic $\frac{h_2(x/\|x\|_2)}{\log n} \approx \frac{h_2(y/\|y\|_2)}{\log k}$.

Combining (D1) and (D2) yields

$$\|Av\|_2^2 \geq \|Mv\|_2^2 \cdot p_w^{-h_2(y)/(2 \log k)}, \quad (2)$$

which is the combined-score lower bound (with the $(s+k)$ denominator of the combined-score formula replaced by its large- k asymptote $2 \log k$).

Upper-bound ingredients. The upper-bound proof swaps in two opposite-direction ingredients:

- (U1) Deterministic intra-block inequality $k_{\text{eff}}(y) \geq 1$, from the power-mean inequality

$$\|y\|_2^4 = \left(\sum_i y_i^2 \right)^2 \geq \sum_i y_i^4 = \|y\|_4^4. \quad (3)$$

Equality iff y is a pure block-spike (support size one).

- (U2) i.i.d. entropy-scaling assumption. Block rows are i.i.d. uniform samples of the n full rows, so

$$\mathbb{E}[\|Mv\|_2^2] = p_w \cdot \|Av\|_2^2. \quad (4)$$

At the participation-ratio level this is equivalent to

$$h_2(x) = h_2(y) + |\log p_w|, \quad (5)$$

i.e. the block collision entropy is lower than the full entropy by exactly $|\log p_w|$. This is the relative-entropy analog of (D2): instead of preserving h_2 as a *fraction* of $\log(\text{dim})$, it preserves h_2 as an *absolute gap* of $|\log p_w|$.

Theorem. Under (U1) and (U2),

$$\|Av\|_2^2 \leq \mathbb{E} \left[\frac{\|Mv\|_2^2 \cdot k_{\text{eff}}(v)}{p_w} \right] = \mathbb{E}[\text{obj}_{\text{UB}}(v)]. \quad (6)$$

Proof. From (U2), after rearranging,

$$\|Av\|_2^2 = \frac{\mathbb{E}[\|Mv\|_2^2]}{p_w}. \quad (\star)$$

From (U1), pointwise $k_{\text{eff}}(y) \geq 1$, so

$$\|Mv\|_2^2 \leq \|Mv\|_2^2 \cdot k_{\text{eff}}(v). \quad (\dagger)$$

Taking expectation of $(\dagger)$ and dividing by p_w ,

$$\frac{\mathbb{E}[\|Mv\|_2^2]}{p_w} \leq \frac{\mathbb{E}[\|Mv\|_2^2 \cdot k_{\text{eff}}(v)]}{p_w} = \mathbb{E}[\text{obj}_{\text{UB}}(v)]. \quad (7)$$

Chaining with $(\star)$ gives $\|Av\|_2^2 \leq \mathbb{E}[\text{obj}_{\text{UB}}(v)]$. ■

Tightness. Both $(\dagger)$ and $(\star)$ are tight exactly when:

- $(\dagger)$ tight: $k_{\text{eff}}(y) = 1$, i.e. y is a block-spike.
- $(\star)$ tight: the i.i.d. unbiasedness $\mathbb{E}[\|Mv\|_2^2] = p_w \|Av\|_2^2$ is exact.

Under both, the single-row spike carries exactly p_w fraction of $\|Av\|_2^2$. Since this is unusual, the upper bound is typically loose; the looseness is concentrated on high- k_{eff} (flat) directions, which is the mechanism by which the argmax of obj_{UB} favors flat directions over spike directions even when the spike direction has larger raw block energy.

Side-by-side with the combined-score lower bound. Both bounds have the form $\|Mv\|_2^2 \cdot \phi(v)$ with $\phi(v) \geq 1$ an entropy multiplier:

- Combined-score LB: $\phi_{\text{LB}}(v) = p_w^{-h_2(y)/(2 \log k)}$.
- UB companion: $\phi_{\text{UB}}(v) = k_{\text{eff}}(y)/p_w = e^{h_2(y)}/p_w$.

Writing both in the form $C \cdot e^{\alpha h_2(y)}$ with C a v -independent constant:

$$\alpha_{\text{LB}} = \frac{|\log p_w|}{2 \log k}, \quad \alpha_{\text{UB}} = 1. \quad (8)$$

The ratio $\alpha_{\text{UB}}/\alpha_{\text{LB}} = 2 \log k / |\log p_w|$; for $k = 32$, $n = 128$, $p_w = 1/4$ this is ≈ 5 , so the UB multiplier is aggressively stronger in h_2 than the LB multiplier.

Alternative: doc-faithful UB under dim-scaled (D2). If we keep doc’s dim-scaled entropy assumption (D2) but also require the i.i.d. L^4 analog $\|x\|_4^4 \leq \mathbb{E}[\|y\|_4^4]/p_w$, then combining (D2)+i.i.d. gives the symmetric UB

$$\|Av\|_2^2 \leq \|Mv\|_2^2 \cdot p_w^{-1/2 - h_2(y)/(2 \log k)}. \quad (9)$$

This UB differs from the combined-score LB only by the v -independent factor $p_w^{-1/2}$, so it preserves the v -ranking of the LB and does not improve separation between flat and spike directions.

The aggressive k_{eff} multiplier of obj_{UB} is only recovered when the entropy-scaling assumption is the i.i.d. absolute-gap form (U2), not the dim-scaled fractional form (D2). The two scalings agree at flat directions ($h_2(x) = \log n$, $h_2(y) = \log k$) but diverge sharply for spike directions, where (D2) predicts $h_2(x) = h_2(y) \log n / \log k = 0$ (since $h_2(y) = 0$) while (U2) predicts $h_2(x) = 0$ as well but with a definite $|\log p_w|$ gap elsewhere — reflecting that i.i.d. sampling concentrates entropy more aggressively than dim-scaling predicts.

Gradient. The objective

$$\text{obj}_{\text{UB}}(v) = \frac{\|Mv\|_2^6}{p_w \|Mv\|_4^4} \quad (10)$$

has $\log \text{obj}_{\text{UB}}(v) = 6 \log \|Mv\|_2 - 4 \log \|Mv\|_4 - \log p_w$, so

$$\nabla \log \text{obj}_{\text{UB}}(v) = 6 \frac{M^\top Mv}{\|Mv\|_2^2} - 4 \frac{M^\top (Mv)^{\odot 3}}{\|Mv\|_4^4}. \quad (11)$$

No new optimizer machinery is required relative to the combined score: drop the c_w scaling on the entropy gradient piece in favor of the fixed coefficients $(6, -4)$ for the $(\log \|\cdot\|_2, \log \|\cdot\|_4)$ terms.

Extension to streaming (pooled operator $C_w = [R; A_w]$). When a carried sketch B and a row memory R are present, the natural streaming form replaces the single block $M = A_w$ with the pooled row-memory operator $C_w = [R; A_w]$ for the entropy factor and the pooled energy operator $M_w = [B; A_w]$ for the energy factor, following the same separation used in the combined score:

$$\text{obj}_{\text{UB},w}(v) := \frac{(\|Bv\|_2^2 + \|A_w v\|_2^2) \cdot k_{\text{eff}}^{(R,w)}(v)}{p_w}, \quad (12)$$

with

$$k_{\text{eff}}^{(R,w)}(v) := \frac{\|C_w v\|_2^4}{\|C_w v\|_4^4}, \quad p_w := \frac{n_{\text{seen}} + k}{n}. \quad (13)$$

The proof carries through: (U1) becomes $k_{\text{eff}}^{(R,w)}(v) \geq 1$ (same power-mean argument on $C_w v$), and (U2) becomes the expectation identity that the carried sketch plus current window form an unbiased energy estimator of the full operator on direction v .